

Inhibitory Control Contributes to “Motor”- but not “Cognitive”- Impulsivity

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Abstract. Literature on impulsivity regularly claims inhibitory control deficits underlie impulsive behavior. The current study investigated whether taxing inhibitory control will increase reflection (decision making under conditions of uncertainty), temporal (delay of gratification), and motor impulsivity (behavioral disinhibition). Inhibitory control was challenged, via a random letter generation task presented during responding to three impulsivity measures: the Information Sampling Task (IST), Single Key Impulsivity Paradigm, and the Stop Signal Task (SST). Participants ($n = 33$) were assigned to the inhibitory control challenging (experimental) condition, or to a control condition in which inhibitory control was not challenged. The SST was affected by the inhibitory control challenge: participants in the experimental condition displayed increased motor impulsivity, evidenced in longer stop signal reaction times (SSRTs) compared to the control group. The manipulation did not affect reflection- or temporal- impulsivity measures. These data support the suggestion that the mechanisms underlying the motor subtype of impulsivity are dissociable from the temporal and reflection subtypes, and that engagement of inhibitory control is not necessary to prevent impulsive decision making.

Keywords: inhibitory control, motor impulsivity, temporal impulsivity, reflection impulsivity, information sampling task, delayed discounting, stop signal task

“Impulsivity” generally refers to a wide range of actions that occur without foresight or thought of future consequences, that are risky and detrimental to the individual (Evenden, 1999) and it represents a central feature of a number of pathologies including drug addiction and gambling (Evenden, 1999; Lawrence, Luty, Bogdan, Sahakian, & Clark, 2009). It has been suggested that impulsivity is a consequence of impaired inhibitory control (e.g., Lawrence et al., 2009; Perales, Verdejo-Garcia, Moya, Lozano, & Perez-Garcia, 2009), a hypothesis for which there is mixed support (Enticott, Ogloff, & Bradshaw, 2006).

Current research suggests that impulsivity is multifactorial and best conceived of as consisting of several subtypes of behavior (Evenden, 1999) reflecting distinctive neurological, pharmacological, and cognitive processes (Enticott et al., 2006; Pattij & Vanderschuren, 2008; Winstanley, Eagle, & Robbins, 2006). A distinction has been proposed between impulsive decision making (or cognitive impulsivity) and impulsive action (Congdon & Canli, 2008). Further from this, three subtypes of behavioral impulsivity have been suggested: “temporal”- and “reflection”- impulsivity, as types of impulsive decision making, and “motor”- impulsivity as impulsive action (Fineberg et al., 2010).

Studies investigating impulsivity originally employed questionnaires of trait impulsiveness, such as the Barratt

Impulsiveness Scale (Patton, Stanford, & Barratt, 1995). However, research suggests questionnaire measures of trait impulsiveness often have little validity as predictors of behaviors, both in real-life and laboratory paradigms (for examples see Butler & Montgomery, 2005; Clark, Robbins, Ersche, & Sahakian, 2006; Lansbergen, Schutter, & Kenemans, 2007; Wingrove & Bond, 1997). Thus, currently, impulsivity research tends to focus on behavioral measures of state impulsivity.

Inability to delay gratification has been referred to as (“temporal”) impulsivity (e.g., Ainslie, 1975; Bickel, Odum, & Madden, 1999; Kirby, Petry, & Bickel, 1999; Richards, Zhang, Mitchell, & de Wit, 1999; Wittmann & Paulus, 2008). This subtype is typically measured with delay discounting questionnaires (e.g., Kirby et al., 1999) or by computerized tasks such as the Single Key Impulsivity Task (SKIP, Dougherty, Marsh-Richard, Hatzis, Nouvion, & Mathias, 2008; Dougherty, Mathias, Marsh, & Jagar, 2005), in which participants are required to choose between small immediate rewards, and large delayed rewards. Individuals high in temporal impulsivity are unable to delay gratification, and will discount delayed rewards more rapidly (Ainslie, 1975).

“Reflection impulsivity” (Kagan, 1965a, 1965b) has been proposed as the tendency to make quick decisions

under conditions of uncertainty (Clark et al., 2006; Clark, Roiser, Robbins, & Sahakian, 2009; DeVito et al., 2009; Messer, 1976). It is challenged by tasks such as the Information Sampling Task (IST, Clark et al., 2006) and Matching Familiar Figures Task (Cairns & Cammock, 1978; Kagan, 1965a), which require participants to gather and monitor information before making a decision. An individual high in impulsivity will make a decision before obtaining all the necessary information.

Finally, “motor impulsivity,” also referred to as “behavioral disinhibition,” refers to the inability to inhibit an already initiated behavior, usually a pre-potent response, when that behavior is no longer appropriate (Chamberlain & Sahakian, 2007; Chamberlain et al., 2007; Ramaekers & Kuypers, 2006; Strakowski et al., 2009; Winstanley et al., 2006). Stop Signal tasks (SSTs) encourage a pre-potent response and then require participants to inhibit this behavior on a small number of trials (Logan, 1994). Disinhibited individuals display slow responding to a stop signal, and are labeled more impulsive (Eagle et al., 2008).

The mechanisms underlying impulsivity are yet to be fully established (Enticott et al., 2006). Research has not explained whether impulsivity can be attributed to decreased top-down processing, or increased bottom-up activation (Perales et al., 2009). It has been hypothesized that impulsivity results from reduced inhibitory control (Enticott et al., 2006; Perales et al., 2009). Some researchers have claimed to have found evidence for the association between inhibitory control and impulsivity in drug user populations. Frontal cortical brain regions associated with inhibitory control are directly affected by long-term exposure to drugs of abuse (Duka et al., 2011; Goldstein & Volkow, 2011; Taylor & Jentsch, 1999) and research finds marked increases in temporal impulsivity (Bjork, Hommer, Grant, & Danube, 2004; Coffey, Gudleski, Saladin, & Brady, 2003; Petry, 2001), reflection impulsivity (Clark et al., 2009; Morgan, Impallomeni, Pirona, & Rogers, 2006), and motor impulsivity (Fillmore & Rush, 2002; Lawrence et al., 2009; Monterosso, Aron, Cordova, Xu, & London, 2005) in drug users.

As with “impulsivity,” there are multiple ways of conceiving and defining inhibitory control (e.g., see Harnishfeger, 1995; Nigg, 2000), but in the context of the present study the term refers to the deliberate, controlled suppression of an explicit pre-potent responses (Enticott et al., 2006; Miyake et al., 2000). In a more general way inhibitory control is considered a key component of executive function (Chikazoe, 2010; Verbruggen & Logan, 2009). Inhibitory control can be invoked to prevent ongoing, motor responses leading to what is known as “behavioral inhibition.” Behavioral inhibition refers to three interrelated processes: (i) inhibition of a pre-potent response to an event; (ii) stopping of an ongoing response, thus permitting a delay in the decision to respond; and (iii) the protection of this period of delay from interference by competing events and responses (“interference control”) (Barkley, 1997). Behavioral inhibition and interference control can be categorized as types of “Executive inhibition” (Nigg, 2000).

When investigating the relationship between impulsivity and inhibitory control, it seems that the concept of motor impulsivity and behavioral inhibition are antipodes, and share definitional features (Bickel, Jarmolowicz, Mueller, Gatchalian, & McClure, 2012). Failure in behavioral inhibition, which would lead to suppression of a pre-potent (inappropriate) response, will result in motor impulsivity. Indeed, the SST is employed both, as a measure of inhibitory control in the executive function field, and as a measure of motor impulsivity (Logan, 1994; Miyake et al., 2000; Ray Li, Yan, Sinha, & Lee, 2008).

However, the assumption that deficits in inhibitory control are responsible for “impulsivity” also implicates cognitive impulsivity. For instance, it has been suggested that behavioral inhibition of an initial response provides a necessary delay in the decision process, allowing successful self-regulation and controlled responding (Barkley, 1997), that is, that inhibitory control allows the suppression of rapid responses and reflexes to allow slower cognitive mechanisms to guide behavior (Taylor & Jentsch, 1999). Studies empirically testing this assumption are few, and the question of whether cognitive impulsive decision making relies on inhibitory control is not yet answered (Perales et al., 2009).

The few studies that have tested the reliance of cognitive impulsivity on inhibitory control have found mixed results. Functional imaging studies show the involvement of prefrontal cortical areas involved in inhibitory control during successful delay of gratification (Diekhof & Gruber, 2010). Petry (2001) has suggested that successful delayed discounting (temporal impulsivity) requires engagement of executive functions, including inhibition, in order to prevent impulsive responding. Another study found that a high working memory load, which challenges inhibition, increased temporal impulsivity, resulting in greater discounting of delayed rewards (Hinson & Whitney, 2006). However, a more recent study has suggested that challenging executive functions does not necessarily increase impulsive decision making, instead simply increases the number of inconsistent choices (Franco-Watkins, Rickard, & Pashler, 2010). The relationship between inhibitory control and reflection impulsivity remains unexplored. It is clear that the assumption that inhibitory control underlies “impulsivity” as a whole needs further exploration. Indeed current research has failed to establish where, and even if, the underlying mechanisms converge between subtypes.

The current study aims to investigate whether cognitive impulsivity, as well as motor impulsivity, is reliant on inhibitory control. It is hypothesized that if successful inhibitory control is necessary to prevent impulsive decision making, challenging such control via a competing dual task should increase impulsive decisions. If motor impulsivity and cognitive impulsivity subtypes are dissociable, as previously suggested, and rely differentially on inhibitory control, then challenging such control should affect the subtypes of impulsivity to different degrees. It was predicted that motor impulsivity, as measured by the SST, should increase when inhibitory control is challenged.

To measure motor-, reflection- and temporal- “impulsivity,” we selected a behavioral task for each of the three subtypes: the SST (Logan, 1994), the IST (Clark et al., 2006), and the SKIP (Dougherty et al., 2005), respectively. Inhibitory control was challenged, during performance on the tasks, using a Random Letter Generation task (RGT, Franco-Watkins et al., 2010). Random generation requires engagement of multiple executive functions, including inhibitory control mechanisms and updating (Fisk & Sharp, 2004; Miyake et al., 2000). “Updating” refers to the dynamic process of updating and monitoring working memory (Miyake et al., 2000). We have assumed that, although updating is a necessary function to monitor responses and check randomness, inhibitory control processes dominate as the most challenging component of the task is to prevent the pre-potent responses to name the letters sequentially (Miyake et al., 2000; Swanson & Sachse-Lee, 2001; Wiegiersma, 1982). To control for any group differences on inhibitory control, baseline measures of inhibitory control on the SST were recorded, along with IQ and memory capacity.

Method

Participants

Thirty-three healthy paid participants were recruited from the University of Sussex subject pool. Participants gave informed consent. The study had been approved by the University of Sussex Ethics Committee.

Participants had been instructed to abstain from the use of illicit recreational drugs for at least 1 week prior to the experiment and from the use of alcohol for at least 12 h prior to the experiment. In addition, participants had been instructed to abstain from drinking coffee or tea in the morning prior to the experiment.

Participants were randomly assigned to either the executive function challenging condition (experimental condition, $N = 17$), or the control condition ($N = 16$). Both conditions involved a dual task running in parallel to the performance in the impulsivity tasks.

Procedure

Participants completed a questionnaire pack including personal details, the Beck Depression Inventory, alcohol and drug use questionnaire, multidimensional mood state questionnaire, and the Barratt’s impulsivity scale. They were then moved to the computer booth where they completed the National Adult Reading Scale and the cognitive tasks at base line, an immediate word recall task, and the SST at baseline. Participants were given the dual task instructions, according to their assigned condition. Participants were informed that they were going to complete three computerized tasks, and were required to complete the dual task alongside the computerized tasks. Participants then

completed the IST, SST, and SKIP, alongside the dual task. The three tasks were completed in a random order. Participants were debriefed before leaving.

Materials

Dual Task

Participants in the experimental condition were required to complete a random letter generation task while performing the impulsivity tasks. Participants were required to generate random letters in response to a tone, every second. In the control condition participants were required to produce a single syllable (“ba”) in response to a tone, every second.

Dependent variables were percentage of errors made during RGT (failing to give a response) and the random number generation (RNG) score (Evans, 1978). The RNG score gives a measure of departure from randomness. It has a range from 0 to 1, where 0 indicates perfect equality of digram distribution, and 1 indicates complete predictability of responses. The RNG score was calculated using RgCalc (Towse & Neil, 1998).

Questionnaires

Personal Details Questionnaire (PDQ)

The PDQ is brief questionnaire asking for age, date of birth, smoking status, alcohol use, and current medication.

Beck Depression Inventory II (BDI-II; Beck, Steer, Ball, & Ranieri, 1996)

The BDI is a 21-item multiple choice checklist measuring severity of clinical depression. On each item participants select which of four statements most applies to them. The questionnaire gives a depression score ranging from “normal” to “extreme depression.” Higher scores indicate worse depression.

Alcohol Use Questionnaire (AUQ; Mehrabian & Russell, 1978)

Participants give an estimation of alcohol drinks consumed per week, and an indication of drinking behavior (for example, number of times drunk in the last year, number of drinks consumed per hour). The AUQ gives a measure of total units per week, binge score, and alcohol use score.

Drug Use Questionnaire (see Townshend & Duka, 2005)

This questionnaire asks for duration of use, time since last use, and how often used for all the main drug categories. For the purposes of this study as a rough guide to drug

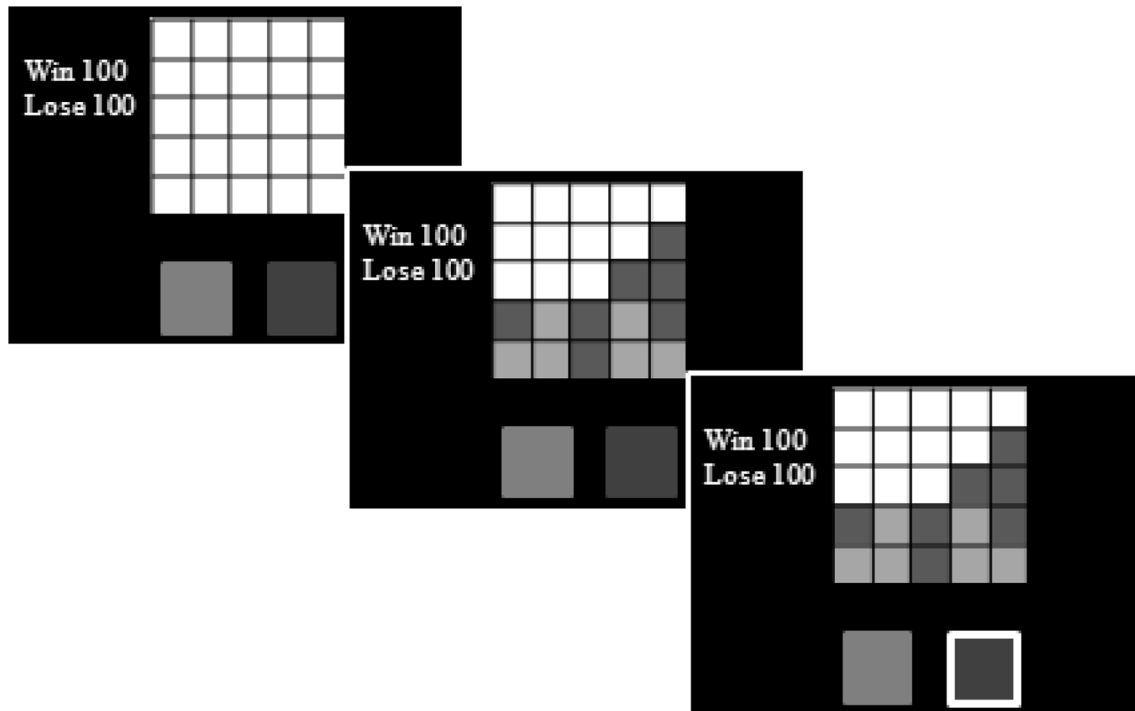


Figure 1. Screen display for the IST. Each trial commences (left) with the presentation of 25 boxes (depicted in figure as white) arranged in a 5 by 5 matrix. Under each box is one of two colors (colors are distributed randomly; colors are depicted in figure as light and dark gray). Participants click on boxes to reveal the color underneath. Participants must decide which of the two colors is in the majority on the board; once they have made their decision they click on the corresponding color at the bottom of the screen (bottom right). Participants are instructed that they can open as many boxes as they wish to make their decision; boxes remain open for the duration of the trial. A feedback response “You have won/lost 100 points” is then presented on the screen.

use, participants were given a score in which 0 = no drug use; 1 = occasional use of cannabis/hash or marijuana; 2 = regular use of cannabis/hash or marijuana (at least once a week); 3 = use of ecstasy and/or other drugs.

Barratt Impulsivity Scale, Version 11 (BIS-11, Patton et al., 1995)

The BIS-11 is a 30-item checklist measuring impulsivity. Participants score on a 4-point scale how much a statement describes them. The questionnaire gives a total impulsivity score, and three subscores: attentional, motor, and non-planning impulsivity.

Multidimensional Mood State Questionnaire (MDMQ; Steyer, Schwenkmezger, Notz & Eid, 1997)

The MDMQ is an English version of the German Multidimensional Mood State Questionnaire. Participants score how much 30 mood states apply to them on a 6-point scale. The questionnaire gives scores on each of the following continuums: good-bad, awake-tired, calm-nervous.

Cognitive Tasks

National Adult Reading Task (NART; Nelson & O’Connell, 1978)

Participants completed the National Adult Reading Task, to give an estimate measure of verbal IQ. The NART requires participants to pronounce a list of 50 irregular words. Number of errors in pronunciation is used as an estimate of verbal IQ.

Rey Auditory-Verbal Learning Test (RAVLT; Rey, 1964)

The immediate word recall task assesses short-term memory. A list of 15 words is read to the participants and after a 1 min delay the participants are asked to verbally recall as many words as they can remember. The task gives an immediate word recall score.

Information Sampling Task (IST; Clark et al., 2006)

The IST assesses Reflection Impulsivity. See Figure 1 for details of the task.

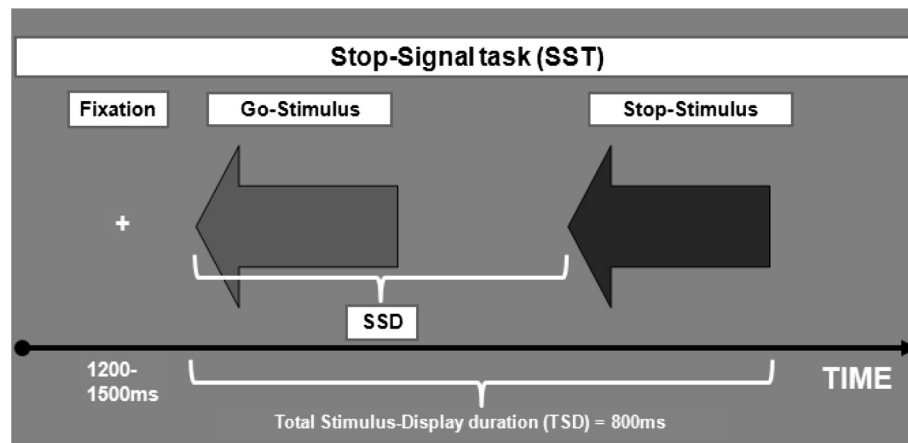


Figure 2. Screen display for an example trial of the SST, with stop signal presentation. A fixation cross is presented, followed by the Go stimulus (a green arrow, depicted in figure as light gray). On 25% of trials the arrow turns from green to red (Stop stimulus, depicted in figure as dark gray). Participants are instructed to respond using button presses to the direction of the green arrow but to withhold this response whenever the Stop Signal is presented. Initial SSD is in 200 ms after the onset of the Go stimulus and is subsequently adjusted using a staircase procedure: when the participant successfully stops to a Stop Signal, the subsequent timing of the Signal is increased by 50 ms; if the participant fails to stop then the subsequent timing is reduced by 50 ms.

Participants completed one practice trial, followed by 10 experimental trials. The dependent variables are the probability of being correct that the individual will tolerate before making a decision ($P(\text{correct})$ see (Clark et al., 2006) for detail), the average number of boxes opened before making a decision, and total number of errors. A large number of errors, few boxes opened, and a lower $P(\text{correct})$ values indicate impulsivity.

Single Key Impulsivity Paradigm (SKIP; Dougherty et al., 2005)

The SKIP assesses ability to delay gratification. Participants are instructed to press the mouse-button to obtain a point reward. The magnitude of the point reward is dependent on the delay between consecutive responses (participants are not informed of this fact). Feedback is displayed on screen for the duration of the task: a point counter at the top of the screen displays the total points accumulated during the session, and a counter at the bottom of the screen displays the number of points earned by the most recent response. The latter point counter provides information regarding the delay-reward contingency, allowing the participant to infer that responses emitted at faster rates earn smaller rewards than responses emitted at slower rates do. The size of the reward is directly proportional to the delay. Participants completed one 4-min trial. The task gives a measure of average inter-response time (the average time delay between consecutive mouse clicks) and total number of responses. A large number of responses, and short IRTs indicate increased impulsivity.

Stop Signal Task (SST; Logan, 1994)

The SST assesses motor impulsivity. See Figure 2 for details of the task.

Participants completed 120 practice trials, 120 trials at baseline, and 120 trials under conditions of the dual task. Twenty-five percent of experimental trials were Stop stimuli. The Stop Signal Reaction Time (SSRT) is the main dependent variable, as the measure of inhibitory control. The task also gives a measure of average Go reaction time (msecs), percentage Go and Stop accuracy, and average Stop Signal Delay (SSD). SSRT can be calculated using the mean (SSRT_m) or the integration (SSRT_i) method: $\text{SSRT}_m = \text{meanGoRT} - \text{mean SSD}$; $\text{SSRT}_i = \text{nth-GoRT} - \text{meanSSD}$ (for a discussion of the two methods, see Verbruggen & Logan, 2009). Large SSRT values indicate poor inhibitory control, that is, increased motor impulsivity.

Statistical Analysis

Baseline group demographics, including the questionnaire measures given, and baseline inhibitory control according to the SST (SSRT_m, SSRT_i), IQ and memory scores, were analyzed using one-way analyses of variance (ANOVAs) to check that the two groups did not differ.

Data from the SST were analyzed using one-way analyses of variance. Condition (control vs. experimental) was applied as the fixed factor, with SSRT_m, SSRT_i, Go accuracy, Go Reaction time, and Stop accuracy applied as the dependent variables.

Table 1. Group characteristics (age, memory, verbal IQ, current mood, alcohol use, inhibitory control) and trait demographics (self-reported impulsivity) for the control and experimental groups

	Experimental	Control	Significance (<i>p</i>)
<i>N</i>	17	16	
Age	21.06(2.68)	20.13(1.26)	.214
RAVLT, words recalled (memory)	7.76(1.95)	8.25(2.35)	.523
NART, estimated verbal IQ	110.94(5.83)	110.08(7.42)	.723
Barratt impulsivity scale			
Total score	65.88(10.89)	69.50(7.96)	.287
Attentional subtype	17.47(3.45)	18.38(3.70)	.473
Motor subtype	23.94(4.39)	25.25(3.44)	.350
Nonplanning subtype	24.47(4.84)	25.88(4.26)	.384
Multidimensional mood state questionnaire			
Good/bad subscale	43.25(8.41)	41.73(7.30)	.597
Awake/tired subscale	36.56(11.25)	38.73(7.32)	.532
Calm/nervous subscale	43.50(8.17)	41.00(6.79)	.364
Alcohol use questionnaire			
Units of alcohol per week	21.13(18.07)	14.29(13.01)	.239
Binge score	22.75(15.02)	26.52(26.19)	.624
AUQ score	46.53(32.62)	40.81(36.13)	.646
SST, baseline inhibitory control			
SSRTm	231.02(41.56)	252.91(83.49)	.347
SSRTi	235.21(45.78)	265.60(80.05)	.191

Note. Values are expressed as mean (*SD*).

Data from the cognitive impulsivity measurements, IST and SKIP, were analyzed for differences between the experimental and control condition using Multivariate Analyses of Variance (MANOVAs). MANOVAs were performed for all the test variables within each task, applying condition (experimental versus control) as a fixed factor. If condition was found to be significant, planned tests of between-subject effects (one-way Analysis of Variance – ANOVA) were performed with the factor condition (experimental versus control) for each variable.

RGT data were analyzed for between impulsivity task differences using repeated-measures ANOVAs. Impulsivity task (IST, SST, SKIP) was included as the repeated-measures factor, with percentage of errors and RNG score as the dependent variables. If the assumption of sphericity was not met, ANOVAs were corrected using Greenhouse-Geisser ($e < .75$) or Huynh-Feldt ($e > .75$). Percentage of errors and RNG score were also correlated with performance on the corresponding impulsivity measure, using Pearson's r , to check for relationships between performance on the RGT and impulsive responding.

To examine the relationship between alcohol use and impulsivity measures, in particular under inhibitory control challenge, correlations were performed between the two scores derived from the alcohol used questionnaire (units drunk per week and binge score) and the impulsive measurements' variables separately for each condition.

To examine the relationship between state and trait impulsivity, correlations between the BIS total and

subscales and the different task variables were also performed.

Results

Group Differences at Baseline

There were no group differences on any baseline measure (see Table 1).

Stop Signal Task

One participant was excluded from the control condition for failing to follow task instructions, seen in a mean Go reaction time of over 1,000 ms.

There were no group differences at baseline on any of the outcome variables (SSRTm, SSRTi), $p > .05$. See Table 1 for descriptive variables.

Under conditions of the dual task there was a significant difference between SSRTm in the control group (264.93 ± 13.76) and in the experimental group (313.53 ± 15.20), $F(1, 30) = 5.503$, $p = .026$; Cohen's $d = .836$, see Figure 3. There was not a significant difference between SSRTi in the control group (270.09 ± 17.18) and in the experimental group (315.06 ± 17.63), $F(1, 30) = 3.299$, $p = .079$; Cohen's $d = .645$.

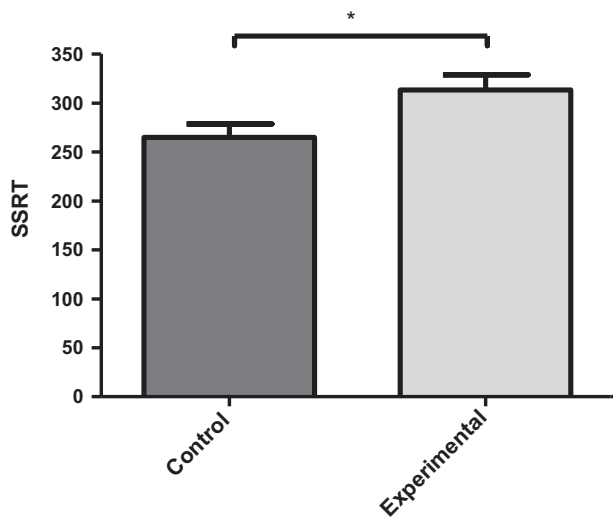


Figure 3. Stop signal reaction times (Mean $\pm$ SEM) on the SST for the control and experimental groups. A higher SSRT indicates greater motor impulsivity. $*p < .05$.

There were no significant differences between groups in Go or Stop accuracy, or Go Reaction Time, $F_s < 2.220$, $p_s > .05$; see Table 2 for descriptive values.

Information Sampling Task

MANOVA with the fixed-factor condition (experimental versus control) on all three test variables showed no significant effect of condition, $F(3, 29) = 1.178$, $p = .335$; $\eta^2 = .109$. See Table 3 for mean values and SDs.

Single Key Impulsivity Task

MANOVA with the fixed-factor condition (experimental versus control) on the two test variables showed no significant effect of condition, $F(2, 30) = .013$, $p = .987$; $\eta^2 = .001$; see Table 3 for descriptive values.

Random Generation Task

Data were missing from three participants. There was no effect of the impulsivity task on the number of errors made on the RGT, $F(1.203, 15.635) = 2.237$, $p = .153$; $\eta^2 = .147$. There was an effect of the task on RNG score, $F(1.721, 20.156) = 7.090$, $p = .006$; $\eta^2 = .353$. Participants displayed less random responding during performance on the SST compared to the SKIP ($p = .001$); there was no difference between the SST and IST, or the IST and SKIP ($p_s > .05$); see Table 4 for descriptive values. There were no significant correlations between errors or RNG scores on the RGT and performance on the corresponding impulsivity measures ($p > .05$).

Table 2. SST key values for the control and experimental groups

	Control				Experimental				Significance (p); Cohen's d
	Mean	Median	SEM	Range	Mean	Median	SEM	Range	
GoRT	596.27	587.88	30.55	434.40–862.54	617.61	626.24	17.62	451.95–730.78	$F(1, 30) = .388$, $p = .538$; .217
%GoRT	619.14	577.83	23.33	420.00–937.07	601.42	619.50	37.95	424.00–798.00	$F(1, 30) = .167$, $p = .686$; .143
GoAcc	93.85	97.78	2.50	61.11–100.00	89.54	91.11	1.59	72.22–100.00	$F(1, 30) = 2.220$, $p = .147$; .521
StopAcc	55.11	53.33	1.25	50.00–63.33	54.51	53.33	1.07	50.00–63.33	$F(1, 30) = .135$, $p = .716$; .129
SSD	331.33	306.67	27.74	193.33–497.06	304.08	326.47	18.64	166.67–391.18	$F(1, 30) = .693$, $p = .412$; .292
SSRTm	264.93	247.31	13.76	172.15–391.49	313.53	303.00	15.20	215.29–459.15	$F(1, 30) = 5.503$, $p = .026$; .836
SSRTi	270.09	257.24	17.18	207.33–466.01	315.06	303.50	17.63	188.53–452.83	$F(1, 30) = 3.299$, $p = .079$; .645

Table 3. SKIP and IST key values for the control and experimental groups

	Experimental	Control	Significance (<i>p</i>); Cohen's <i>d</i>
<i>N</i>	16	17	
Single key impulsivity paradigm			
Total number of responses	239.12 ± 81.24	256.25 ± 70.66	$F(1, 31) = .025, p = .875; -.120$
Average interresponse time	5.32 ± 1.55	5.24 ± 2.18	$F(1, 31) = .001, p = .976; .010$
Information sampling task			
Number of boxes opened	18.02 ± 1.12	17.82 ± 1.05	$F(1, 31) = .018, p = .895; .045$
<i>P</i> (correct)	.8811 ± .0215	.8838 ± .0202	$F(1, 31) = .008, p = .928; .000$
Number of Errors	1.71 ± .24	1.06 ± .23	$F(1, 31) = 3.715, p = .063; .676$

Note. Values are expressed as mean ± SEM.

Correlational Analysis

There were no significant correlations between the Barratt Impulsiveness Scales and any impulsivity tasks ($p > .05$).

Discussion

The current study provided evidence that motor impulsivity is dependent on inhibitory control; challenging inhibitory control processes did not increase impulsive decision making, on either the temporal or reflection subtypes. These findings support the suggestion that impulsivity is not a unitary construct, and should be conceived of as consisting of dissociable subtypes: motor impulsivity appears to be, at least partly, distinct from cognitive reflection- and temporal- impulsivity. Inhibitory control was challenged by introducing as a dual task, the random letter generation task, which requires inhibition of the pre-potent response to name the letters sequentially rather than randomly.

The results showed that challenging inhibitory control increased impulsive responding on the SST, with no impairment on responses to Go stimuli. Participants in the experimental condition showed less successful inhibition to a stop signal, according to SSRTm values, suggesting that motor impulsivity is dependent on inhibitory control. This is in agreement with previous research which has suggested motor impulsivity and inhibitory control are antipodes (Bickel et al., 2012). The present findings are also consistent with findings in cocaine users showing impairment in inhibiting, but not executing, a go response (Fillmore & Rush, 2002). It is noteworthy that random generation does not uniquely tap inhibitory control processes, it

also invokes other cognitive functions including updating (Swann, Lijffijt, Lane, Steinberg, & Moeller, 2009). We therefore cannot claim that motor impulsivity is dependent *only* on inhibitory control, as it may be that the challenge to updating processes was partly responsible for the increases in motor impulsivity. Thus the current study suggests a need for further research to clarify whether other mechanisms are also involved in the generation of motor impulsivity.

There were differences between SSRT values when calculated using the mean compared to the integration method. There was a significant difference with a large effect size between SSRTs when calculated using the mean method, whereas this difference only approached significance using the integration method. The integration method is applicable if the GO accuracy differs between groups, or if Stop accuracy differs from 50%. In the present study there were no group differences in Stop accuracy, however both groups had approximately 55% Stop accuracy. This high Stop accuracy suggests that the integration method may be a more appropriate calculation. There was variability in Go reaction times, resulting in additional variance when SSRT was calculated using the integration rather than the mean method, which probably led to the loss of a significant effect ($p = 0.07$). Nevertheless there was still a strong effect size.

The lack of an effect of the experimental condition on either temporal- or reflection- impulsivity performance suggests that the temporal- and reflection- subtypes of cognitive impulsivity are distinct from motor impulsivity, and are not reliant on the functions that random generation taps. The current study found no evidence for increased impulsive decision making (cognitive impulsivity) on either subtype when inhibitory control (and updating) was challenged. This supports the proposed distinction between

Table 4. Random generation during performance on the SST, IST, and the SKIP

	SST	IST	SKIP	Significance (<i>p</i>)
Random generation task				
Errors	5.500 ± 2.284	10.286 ± 2.756	11.071 ± 3.697	$F(1.203, 15.635) = 2.237, p = .153$
RNG	.206 ± .009	.193 ± .015	.174 ± .009	$F(1.721, 20.156) = 7.090, p = .006$

Note. Values are expressed as mean ± SEM.

impulsive action and impulsive decision making, with impulsive action arising from impaired inhibitory control (Field, Schoenmakers, & Wiers, 2008; Mitchell, Fields, D’Esposito, & Boettiger, 2005).

The finding that challenging inhibitory control does not increase cognitive impulsivity is interesting. It has previously been suggested that successful inhibitory control of an initial response is required to allow slower cognitive mechanisms to guide behavior (Barkley, 1997; Groman, James, & Jentsch, 2009). Previous research has suggested successful delayed discounting may be reliant on engaged inhibitory control (Diekhof & Gruber, 2010; Petry, 2001). Hinson and Whitney found that a high working memory load, which challenges inhibition, increased temporal impulsivity, seen in greater discounting of delayed rewards. However, our results support recent findings that taxing executive processes, via a random generation task, does not increase impulsive decision making (Franco-Watkins et al., 2010). Taken these findings together it becomes clear that there is need for more research into the mechanisms that underlie the different types of impulsivity. Bickel and colleagues (Bickel et al., 2012) have suggested that reflection- and temporal- impulsivity are antipodal to planning and valuing future events, and not to inhibitory control, a suggestion that merits future research.

There were no differences in generation of letters during the dual task running alongside the different impulsivity measures. Random performance was slightly impaired while running in parallel to the SST, but showed no impairments when performed in parallel to the SKIP or IST, presumably because inhibitory control mechanisms were being challenged by the SST and the RGT simultaneously. The results can lend support to our suggestion that cognitive impulsivity may not be reliant on the inhibitory control mechanisms challenged by random generation, as participants showed relatively random performance on the dual task but no deficits in the cognitive impulsivity measures.

There were limitations to the current study that should be considered. It is possible that the results found reflect differences in difficulty of the three impulsivity measures. The IST was designed as a measure of reflection impulsivity that does not place high demands on visual search and visual working memory (Clark et al., 2006). The SKIP is designed to be simple measure of ability to delay gratification, requiring minimal attentional resources (Swann et al., 2009). The SST is, perhaps, more difficult, and requires cognitive processes for the evaluation of the task demands in addition to processes for response inhibition. The SST is itself a dual task, requiring preparation of two responses (Go and Stop) (Verbruggen, Adams, & Chambers, 2012); it requires engagement of attentional resources, continued monitoring of the stop signal is necessary in order to initiate response inhibition, and maintenance of attention is required for both correct GO and STOP responses (Ray Li et al., 2008). The experimental condition did not affect measurements related to go responses, instead affected only stop responses, which involve inhibitory control. It is possible that this later finding is due to different difficulty in performing the two responses (where Go stimuli require a pre-potent, “easy” response).

The current study suggests that we should consider motor impulsivity to be distinct from cognitive impulsivity. The results highlight the need for more rigorous definitions of impulsivity (Enticott et al., 2006; Field et al., 2008) and more specificity in the constructs that have been referred to as “impulsivity.” If we are to label all these constructs “impulsivity” (as the literature does) then we need more rigorous testing of the mechanisms that underlie them. Further studies are required to characterize all the facets of behavioral impulsivity.

In conclusion, we have shown that a dual task, which challenges predominantly inhibitory control, increased motor impulsivity but had no effect on reflection- or temporal- impulsivity. These findings provide support for the distinction between impulsive action and impulsive decision making, and support the suggestion that taxing inhibitory control processes does not necessarily increase impulsive decision making.

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References

- Ainslie, G. (1975). Specious reward: A behavioral theory of impulsiveness and impulse control. *Psychological Bulletin*, 82, 463–496.
- Barkley, R. A. (1997). Behavioral inhibition, sustained attention, and executive functions: Constructing a unifying theory of ADHD. *Psychological Bulletin*, 121, 65–94.
- Beck, A. T., Steer, R. A., Ball, R., & Ranieri, W. F. (1996). Comparison of beck depression inventories-IA and-II in psychiatric outpatients. *Journal of Personality Assessment*, 67, 588–597.
- Bickel, W. K., Jarmolowicz, D. P., Mueller, E. T., Gatchalian, K. M., & McClure, S. M. (2012). Are executive function and impulsivity antipodes? A conceptual reconstruction with special reference to addiction. *Psychopharmacology (Berl)*, 221, 361–387.
- Bickel, W. K., Odum, A. L., & Madden, G. J. (1999). Impulsivity and cigarette smoking: Delay discounting in current, never, and ex-smokers. *Psychopharmacology (Berl)*, 146, 447–454.
- Bjork, J. M., Hommer, D. W., Grant, S. J., & Danube, C. (2004). Impulsivity in abstinent alcohol-dependent patients: Relation to control subjects and type 1/type 2-like traits. *Alcohol*, 34, 133–150.
- Butler, G. K., & Montgomery, A. M. (2005). Subjective self-control and behavioural impulsivity coexist in anorexia nervosa. *Eating Behaviors*, 6, 221–227.
- Cairns, E., & Cammock, T. (1978). Development of a more reliable version of the matching familiar figures test. *Developmental Psychology*, 14, 555–560.
- Chamberlain, S. R., Fineberg, N. A., Menzies, L. A., Blackwell, A. D., Bullmore, E. T., Robbins, T. W., & Sahakian, B. J.

- (2007). Impaired cognitive flexibility and motor inhibition in unaffected first-degree relatives of patients with obsessive-compulsive disorder. *American Journal of Psychiatry*, 164, 335–338.
- Chamberlain, S. R., & Sahakian, B. J. (2007). The neuropsychiatry of impulsivity. *Current Opinion in Psychiatry*, 20, 255–261.
- Chikazoe, J. (2010). Localizing performance of go/no-go tasks to prefrontal cortical subregions. *Current Opinion in Psychiatry*, 23, 267–272.
- Clark, L., Robbins, T. W., Ersche, K. D., & Sahakian, B. J. (2006). Reflection impulsivity in current and former substance users. *Biological Psychiatry*, 60, 515–522.
- Clark, L., Roiser, J. P., Robbins, T. W., & Sahakian, B. J. (2009). Disrupted “reflection” impulsivity in cannabis users but not current or former ecstasy users. *Journal of Psychopharmacology*, 23, 14–22.
- Coffey, S. F., Gudleski, G. D., Saladin, M. E., & Brady, K. T. (2003). Impulsivity and rapid discounting of delayed hypothetical rewards in cocaine-dependent individuals. *Experimental and Clinical Psychopharmacology*, 11, 18–25.
- Congdon, E., & Canli, T. (2008). A neurogenetic approach to impulsivity. *Journal of Personality*, 76, 1447–1484.
- DeVito, E. E., Blackwell, A. D., Clark, L., Kent, L., Dezser, A. M., Turner, D. C., ... Sahakian, B. J. (2009). Methylphenidate improves response inhibition but not reflection-impulsivity in children with attention deficit hyperactivity disorder (ADHD). *Psychopharmacology (Berl)*, 202, 531–539.
- Diekhof, E. K., & Gruber, O. (2010). When desire collides with reason: Functional interactions between anteroventral prefrontal cortex and nucleus accumbens underlie the human ability to resist impulsive desires. *The Journal of Neuroscience*, 30, 1488–1493.
- Dougherty, D. M., Marsh-Richard, D. M., Hatzis, E. S., Novion, S. O., & Mathias, C. W. (2008). A test of alcohol dose effects on multiple behavioral measures of impulsivity. *Drug Alcohol Dependence*, 96, 111–120.
- Dougherty, D. M., Mathias, C. W., Marsh, D. M., & Jagar, A. A. (2005). Laboratory behavioral measures of impulsivity. *Behavior Research Methods*, 37, 82–90.
- Duka, T., Trick, L., Nikolaou, K., Gray, M. A., Kempton, M. J., Williams, H., ... Stephens, D. N. (2011). Unique brain areas associated with abstinence control are damaged in multiply detoxified alcoholics. *Biological Psychiatry*, 70, 545–552.
- Eagle, D. M., Baunez, C., Hutcheson, D. M., Lehmann, O., Shah, A. P., & Robbins, T. W. (2008). Stop-signal reaction-time task performance: Role of prefrontal cortex and subthalamic nucleus. *Cerebral Cortex*, 18, 178–188.
- Enticott, P. G., Ogloff, J. R. P., & Bradshaw, J. L. (2006). Associations between laboratory measures of executive inhibitory control and self-reported impulsivity. *Personality and Individual Differences*, 41, 285–294. doi: 10.1016/j.paid.2006.01.011
- Evans, F. J. (1978). Monitoring attention deployment by random number generation – index to measure subjective randomness. *Bulletin of the Psychonomic Society*, 12, 35–38.
- Evenden, J. L. (1999). Varieties of impulsivity. *Psychopharmacology (Berl)*, 146, 348–361.
- Field, M., Schoenmakers, T., & Wiers, R. W. (2008). Cognitive processes in alcohol binges: A review and research agenda. *Current Drug Abuse Reviews*, 1, 263–279.
- Fillmore, M. T., & Rush, C. R. (2002). Impaired inhibitory control of behavior in chronic cocaine users. *Drug and Alcohol Dependence*, 66, 265–273. doi: 10.1016/S0376-8716(01)00206-X
- Fineberg, N. A., Potenza, M. N., Chamberlain, S. R., Berlin, H. A., Menzies, L., Bechara, A., ... Hollander, E. (2010). Probing compulsive and impulsive behaviors, from animal models to endophenotypes: A narrative review. *Neuropsychopharmacology*, 35, 591–604.
- Fisk, J. E., & Sharp, C. A. (2004). Age-related impairment in executive functioning: Updating, inhibition, shifting, and access. *Journal of Clinical and Experimental Neuropsychology*, 26, 874–890.
- Franco-Watkins, A. M., Rickard, T. C., & Pashler, H. (2010). Taxing executive processes does not necessarily increase impulsive decision making. *Experimental Psychology*, 57, 193–201.
- Goldstein, R. Z., & Volkow, N. D. (2011). Dysfunction of the prefrontal cortex in addiction: Neuroimaging findings and clinical implications. *Nature Reviews Neuroscience*, 12, 652–669.
- Groman, S. M., James, A. S., & Jentsch, J. D. (2009). Poor response inhibition: At the nexus between substance abuse and attention deficit/hyperactivity disorder. *Neuroscience and Biobehavioral Reviews*, 33, 690–698.
- Harnishfeger, K. (1995). The development of cognitive inhibition: Theories, definitions, and research evidence. In F. N. Dempster & C. J. Brainerd (Eds.), *Interference and Inhibition in Cognition* (pp. 175–204). San Diego, CA, US: Academic Press.
- Hinson, J. M., & Whitney, P. (2006). Working memory load and decision making: A reply to Franco-Watkins, Pashler, and Rickard (2006). *Journal of Experimental Psychology: Learning Memory and Cognition*, 32, 448–450.
- Kagan, J. (1965a). Individual differences in the resolution of response uncertainty. *Journal of Personality and Social Psychology*, 56, 154–160.
- Kagan, J. (1965b). Reflection-impulsivity and reading ability in primary grade children. *Child Development*, 36, 609–628.
- Kirby, K. N., Petry, N. M., & Bickel, W. K. (1999). Heroin addicts have higher discount rates for delayed rewards than non-drug-using controls. *Journal of Experimental Psychology*, 128, 78–87.
- Lansbergen, M. M., Schutter, D. J., & Kenemans, J. L. (2007). Subjective impulsivity and baseline EEG in relation to stopping performance. *Brain Research*, 1148, 161–169.
- Lawrence, A. J., Luty, J., Bogdan, N. A., Sahakian, B. J., & Clark, L. (2009). Impulsivity and response inhibition in alcohol dependence and problem gambling. *Psychopharmacology (Berl)*, 207, 163–172.
- Logan, G. D. (1994). On the ability to inhibit thought and action: A users’ guide to the stop signal paradigm. In D. Dagenbach & T. H. Carr (Eds.), *Inhibitory processes in attention memory and language* (pp. 189–239). San Diego, CA: Academic Press.
- Mehrabian, A., & Russell, J. A. (1978). A questionnaire measure of habitual alcohol use. *Psychological Reports*, 43, 803–806.
- Messer, S. B. (1976). Reflection-impulsivity – review. *Psychological Bulletin*, 83, 1026–1052.
- Mitchell, J. M., Fields, H. L., D’Esposito, M., & Boettiger, C. A. (2005). Impulsive responding in alcoholics. *Alcoholism, Clinical and Experimental Research*, 29, 2158–2169.
- Miyake, A., Friedman, N. P., Emerson, M. J., Witzki, A. H., Howerter, A., & Wager, T. D. (2000). The unity and diversity of executive functions and their contributions to complex “Frontal Lobe” tasks: A latent variable analysis. *Cognitive Psychology*, 41, 49–100.
- Monterosso, J. R., Aron, A. R., Cordova, X., Xu, J. S., & London, E. D. (2005). Deficits in response inhibition associated with chronic methamphetamine abuse. *Drug Alcohol Dependence*, 79, 273–277.
- Morgan, M. J., Impallomeni, L. C., Pirona, A., & Rogers, R. D. (2006). Elevated impulsivity and impaired decision-making in abstinent Ecstasy (MDMA) users compared to polydrug

- and drug-naïve controls. *Neuropsychopharmacology*, 31, 1562–1573.
- Nelson, H. E., & O'Connell, A. (1978). Dementia: The estimation of premorbid intelligence levels using the New Adult Reading Test. *Cortex*, 14, 234–244.
- Nigg, J. T. (2000). On inhibition/disinhibition in developmental psychopathology: Views from cognitive and personality psychology and a working inhibition taxonomy. *Psychological Bulletin*, 126, 220–246.
- Pattij, T., & Vanderschuren, L. J. (2008). The neuropharmacology of impulsive behaviour. *Trends in Pharmacological Sciences*, 29, 192–199.
- Patton, J. H., Stanford, M. S., & Barratt, E. S. (1995). Factor structure of the Barratt impulsiveness scale. *Journal of Clinical Psychology*, 51, 768–774.
- Perales, J. C., Verdejo-Garcia, A., Moya, M., Lozano, O., & Perez-Garcia, M. (2009). Bright and dark sides of impulsivity: Performance of women with high and low trait impulsivity on neuropsychological tasks. *Journal of Clinical and Experimental Neuropsychology*, 31, 927–944.
- Petry, N. M. (2001). Delay discounting of money and alcohol in actively using alcoholics, currently abstinent alcoholics, and controls. *Psychopharmacology (Berl)*, 154, 243–250.
- Ramaekers, J. G., & Kuypers, K. P. (2006). Acute effects of 3, 4-methylenedioxymethamphetamine (MDMA) on behavioral measures of impulsivity: Alone and in combination with alcohol. *Neuropsychopharmacology*, 31, 1048–1055.
- Ray Li, C.-S., Yan, P., Sinha, R., & Lee, T.-W. (2008). Subcortical processes of motor response inhibition during a stop signal task. *Neuroimage*, 41, 1352–1363. doi: 10.1016/j.neuroimage.2008.04.023
- Rey, A. (1964). *L'examen clinique en psychologie*. Paris: Presses universitaires de France.
- Richards, J. B., Zhang, L., Mitchell, S. H., & de Wit, H. (1999). Delay or probability discounting in a model of impulsive behavior: Effect of alcohol. *Journal of the Experimental Analysis of Behavior*, 71, 121–143.
- Steyer, R., Schwenkmezger, P., Notz, P., & Eid, M. (1997). *Der Mehrdimensionale Befindlichkeitsfragebogen (MDBF)* [The multidimensional mood state questionnaire (MDMQ)]. Göttingen, Germany: Hogrefe. Retrieved from <http://www.metheval.uni-jena.de/mdbf.php>.
- Strakowski, S. M., Fleck, D. E., DelBello, M. P., Adler, C. M., Shear, P. K., McElroy, S. L., ... Arndt, S. (2009). Characterizing impulsivity in mania. *Bipolar Disorders*, 11, 41–51.
- Swann, A. C., Lijffijt, M., Lane, S. D., Steinberg, J. L., & Moeller, F. G. (2009). Trait impulsivity and response inhibition in antisocial personality disorder. *Journal of Psychiatric Research*, 43, 1057–1063.
- Swanson, H. L., & Sachse-Lee, C. (2001). Mathematical problem solving and working memory in children with learning disabilities: Both executive and phonological processes are important. *Journal of Experimental Child Psychology*, 79, 294–321.
- Taylor, J. R., & Jentsch, J. D. (1999). Impulsivity resulting from frontostriatal dysfunction in drug abuse: Implications for the control of behavior by reward-related stimuli. *Psychopharmacology (Berl)*, 146, 373–390.
- Townshend, J. M., & Duka, T. (2005). Binge drinking, cognitive performance and mood in a population of young social drinkers. *Alcoholism, Clinical and Experimental Research*, 29, 317–325.
- Towse, J. N., & Neil, D. (1998). Analyzing human random generation behavior: A review of methods used and a computer program for describing performance. *Behavior Research Methods Instruments & Computers*, 30, 583–591.
- Verbruggen, F., Adams, R., & Chambers, C. D. (2012). Proactive motor control reduces monetary risk taking in gambling. *Psychological Science*, 23, 805–815.
- Verbruggen, F., & Logan, G. D. (2009). Models of response inhibition in the stop-signal and stop-change paradigms. *Neuroscience and Biobehavioral Reviews*, 33, 647–661.
- Wiegersma, S. (1982). A control-theory of sequential response production. *Psychological Research*, 44, 175–188.
- Wingrove, J., & Bond, A. J. (1997). Impulsivity: A state as well as trait variable. Does mood awareness explain low correlations between trait and behavioural measures of impulsivity? *Personality Individual Difference*, 22, 333–339.
- Winstanley, C. A., Eagle, D. M., & Robbins, T. W. (2006). Behavioral models of impulsivity in relation to ADHD: Translation between clinical and preclinical studies. *Clinical Psychology Review*, 26, 379–395.
- Wittmann, M., & Paulus, M. P. (2008). Decision making, impulsivity and time perception. *Trends in Cognitive Sciences*, 12, 7–12.

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